University of Tehran

Farabi College

Advanced Online Business Course

Stage 1: Project Idea Submission

Arbaeen Pilgrim Assistant Chatbot

Mohammed Razzaq Mohsen Al-Tayyar
Student ID: 220704109

Arbaeen Pilgrim Assistant Chatbot

1. The Main Problem

Arbaeen is one of the largest annual human gatherings on earth, drawing an estimated 20 to 25 million pilgrims to Karbala, Iraq each year. Pilgrims travel from across Iraq and from dozens of countries worldwide, many of them walking the approximately 80-kilometer route from Najaf to Karbala on foot. This immense scale creates serious and recurring practical challenges. Pilgrims frequently become separated from their groups, struggle to locate Mawkibs (the volunteer-run food and rest stations distributed along the route), suffer from heat exhaustion and dehydration under Iraq's intense temperatures, and face significant language barriers given the international composition of the crowd. Access to timely, accurate, and multilingual guidance during the walk remains a genuine and unresolved problem for millions of participants each year.

2. Target Users

We aim to serve pilgrims attending the Arbaeen walk who need quick and reliable answers during their journey. The primary users are Iraqi and international pilgrims, many of whom carry smartphones but do not have access to organized information services while on the road. Furthermore, volunteers stationed along the route and organizers of Mawkib points can also benefit from the system as a tool to guide pilgrims who approach them with questions. The system is designed to be simple enough for users with limited technical experience, since the majority of pilgrims are not technology specialists and require a conversational and intuitive interface.

3. Why This Problem Matters

Arbaeen represents a unique logistical and humanitarian challenge that recurs every year at a massive scale within Iraq. The event carries deep religious, cultural, and national significance for millions of people. Moreover, the challenges pilgrims face during the walk, including medical emergencies, disorientation, and lack of information, are well documented and directly affect the safety and wellbeing of participants. We believe that even a modest improvement in information accessibility during the walk can have a meaningful impact on the experience and safety of a large number of people. This project therefore addresses a real, locally grounded, and socially important problem rather than a hypothetical one.

4. The Proposed Solution

We propose to build a web-based conversational chatbot specifically designed to assist pilgrims during the Arbaeen walk. The chatbot will use artificial intelligence to respond to natural language queries in a helpful and accessible manner. It will focus on three core functions. First, it will provide guidance on locating Mawkib stations and rest points along the Najaf to Karbala route. Second, it will offer basic first aid advice for the most common health issues pilgrims encounter, including heat exhaustion, dehydration, and foot injuries. Third, it will answer general questions about the route, conduct, and logistics of the Arbaeen walk.

Furthermore, the interface will be designed with a mobile-first layout to ensure smooth usability on smartphones, which are the primary devices pilgrims carry. The AI model will operate under a carefully designed system prompt that limits responses to Arbaeen-relevant topics, ensuring focused and responsible outputs.

5. Technologies and AI Tools

Component	**Tool and Technology**
Frontend	React.js, responsive and mobile-first
AI Model	Llama 3.1 8B via Groq API (free tier)
Hosting	Vercel or GitHub Pages (free)
Version Control	Git and GitHub
Development AI Assistance	ChatGPT, Claude, GitHub Copilot

The Groq API provides access to powerful open-source language models at no cost, which makes it well suited for an academic project with no dedicated budget. The REST-based interface integrates cleanly with a React frontend and does not require a separate backend server, keeping the architecture simple and maintainable within the scope of a course project.

6. Initial Project Timeline

Week	**Sprint**	**Deliverable**
1	Sprint 0	Idea approval, environment setup, Groq API integration test
2	Sprint 1	Core chatbot UI, Arbaeen-specific system prompt, basic conversation flow
3	Sprint 2	Mawkib directory, first aid module, mobile responsiveness, user testing
4	Sprint 3	Feedback integration, performance refinement, MVP freeze
5 and 6	Polish	Documentation, poster, final presentation preparation